Nursing home quality from lakehouse to insight

Architecture, dimensional modeling, pipeline execution, Power BI, and FinOps.

09	04	1.36
managed Gold tables	dependent Job tasks	observed Job DBU

Executive summary

CareWatch transforms public CMS nursing-home data through Landing, Bronze, and Silver into a Gold dimensional model. A scheduled Databricks Job completed on 25 October 2026. The Gold task reconciled 698,750 Silver rows and made no Gold mutations on unchanged input.

This report separates measured Job usage from annual planning estimates. Its Power BI and Spark screenshots are direct evidence, while citation, fine, staffing, and rating visuals remain descriptive screening tools.

25 OCTOBER 2026 / AWS US EAST (OHIO) / DATABRICKS + POWER BI

A governed path from CMS to Power BI

The business data moves through four distinct layers. Operational logging and quarantine run alongside the analytical path.

01 CMS source	Public provider, health deficiency, penalty, and MDS publications.
02 Landing	Source manifest, checksum, publication overlap, and acquisition watermark.
03 Bronze	Raw string records with batch and file lineage; source shape retained.
04 Silver	Typed, deduplicated, validated records with idempotent updates.
05 Gold	Conformed dimensions, event facts, and a state-month aggregate.
06 Power BI	Databricks SQL Warehouse and a one-to-many semantic model.

Control plane

The 03:00 UTC daily Job enforces acquire_cms → raw_to_bronze → bronze_to_silver → silver_to_gold. Validation, quarantine, execution audit, and row-level reconciliation determine whether BI should refresh.

Source: deployment/carewatch_daily_incremental.job.json; Job 1053304552502201.

The implemented semantic model

The Power BI Model view confirms dimensions filter facts in one direction. Monitoring views are separate from care-quality facts.

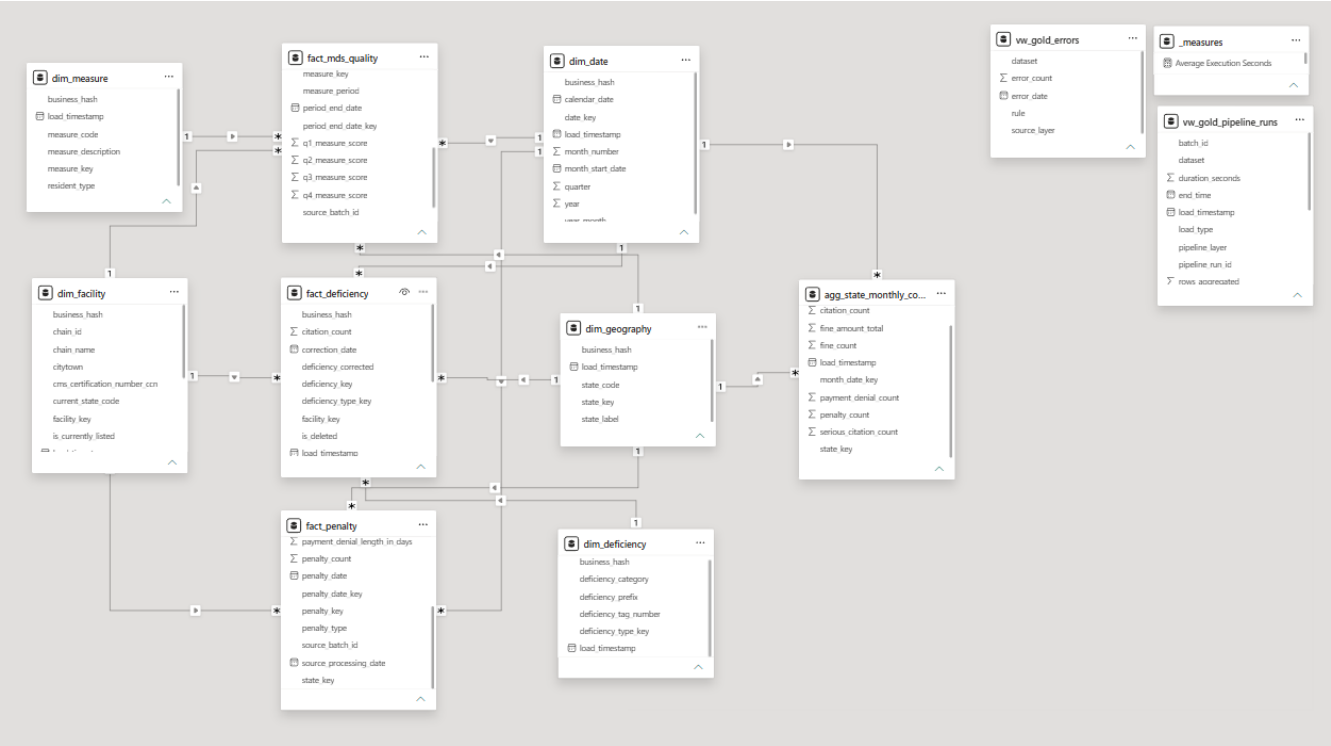


Figure 1. User-supplied Power BI Model view. The image is embedded at its original resolution for zooming.

One-side dimension	Many-side target and key
dim_facility	fact_deficiency, fact_penalty, fact_mds_quality via facility_key
dim_date	Three event facts via their date keys; state-month aggregate via month_date_key
dim_geography	Three event facts and state-month aggregate via state_key
dim_deficiency / dim_measure	Deficiency and MDS facts via their respective type/measure keys

vw_gold_pipeline_runs, vw_gold_errors, and _measures have no physical relationship to business facts. Their independence is intentional.

Gold grains and metric contracts

Five dimensions, three detailed facts, and one aggregate serve reporting without copying Silver tables.

Object	Analytical grain	Rows
fact_deficiency	One citation (deficiency_key)	418,911
fact_penalty	One penalty event (penalty_key)	15,419
fact_mds_quality	One facility × measure × period (mds_key)	249,730
agg_state_monthly_compliance	One state × month (state_key, month_date_key)	3,112

Dimensions

Dimension	Rows	Purpose
dim_facility	14,690	Current provider profile and capacity
dim_date	3,466	Calendar and event-period attributes
dim_geography	54	Observed state and unknown member
dim_deficiency	263	Citation classification
dim_measure	17	MDS measure metadata

Reporting rules

Serious citations use severity G–I or J–L and exclude soft-deleted facts. The current 12-complete-month window is 1 October 2025 through 30 September 2026. Facility rating and ownership describe the current profile, not historical event state.

Source: docs/gold_data_dictionary.md; live Gold counts, 25 October 2026.

A successful scheduled end-to-end run

The Job chain succeeded on 25 October 2026. The screenshot is the scheduled run timeline, not a manual notebook test.

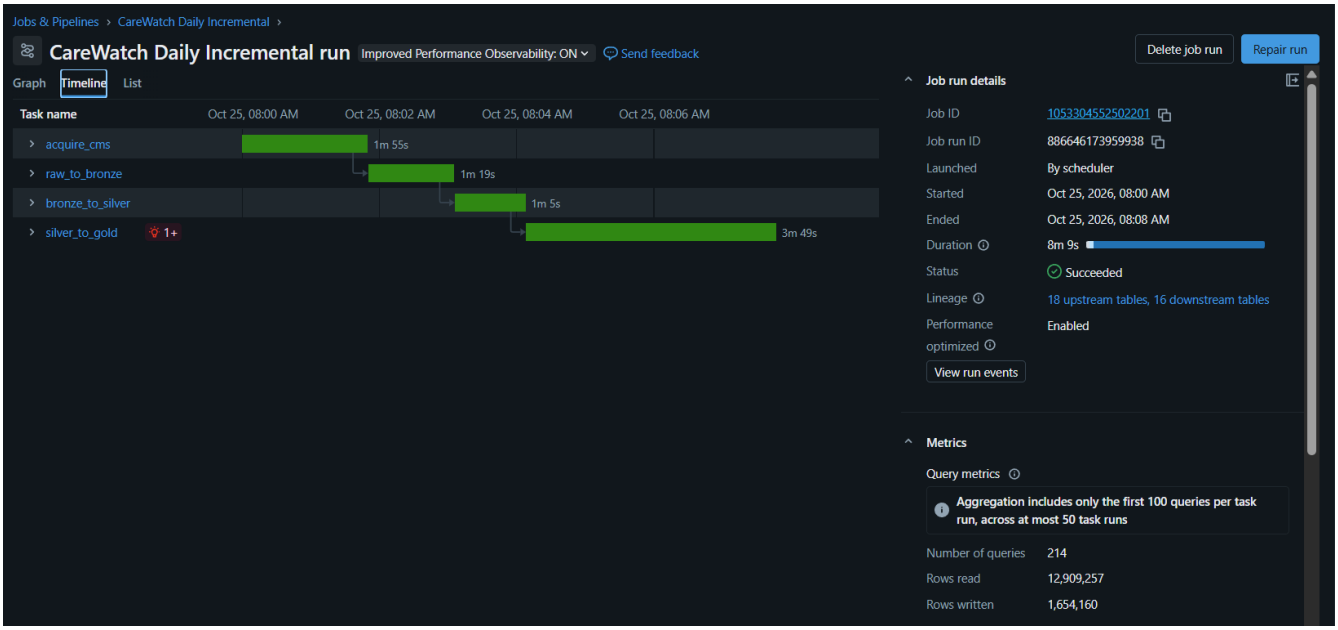


Figure 2. Job run 886646173959938; four dependent tasks finished in 489.320 seconds (8 minutes 9 seconds).

698,750
Silver rows read by Gold

434,330
active event rows aggregated

0
Gold row mutations

What the run proves

The final Gold audit record has status SUCCESS. All table merges and validations completed, and an unchanged input produced zero inserts, updates, or deletes. This run supports scheduled chaining and retry-safe current state; it does not by itself demonstrate a newly changed fact reaching Gold.

The Gold audit reports duration_seconds = 217. The Job task timer is about 227 seconds; these measure slightly different boundaries.

Source: Databricks Job timeline and carewatch.pipeline.vw_gold_pipeline_runs.

Design choices and boundaries

Each choice follows the observed data volume, source behavior, or recovery contract.

Decision	Reason and consequence
Unpartitioned managed Delta	Nine Gold tables use 51.78 MiB of active data files. Partitioning would create small files without a measured query benefit.
No routine OPTIMIZE	A trial skipped the single 30.3 MB deficiency file and rewrote nothing. Reassess when file counts and query latency justify it.
Schema drift and quarantine	Bronze preserves source shape and logs changes. Silver safe-casts typed fields; invalid rows enter restricted quarantine.
Deterministic MERGE	Stable keys and business hashes make retries idempotent. The facility dimension is current-state SCD1.
Privacy boundary	Clear-text contact fields are removed; addresses are salted-hashed in Silver. Gold error views expose counts, not raw payloads.
Repair after failure	Gold targets commit separately. Retry the task after a partial failure and refresh Power BI only after all targets validate.

Source-date interpretation

State on an event comes from the event fact. Current state and ownership on dim_facility describe today's provider profile. Analyses that mix them must name the difference.

Source: Phase 2 evidence, Gold data dictionary, deployment test, and OPTIMIZE result.

A Gold query-profile DAG

This image is an actual Databricks operator graph from a Gold MERGE during the scheduled run.

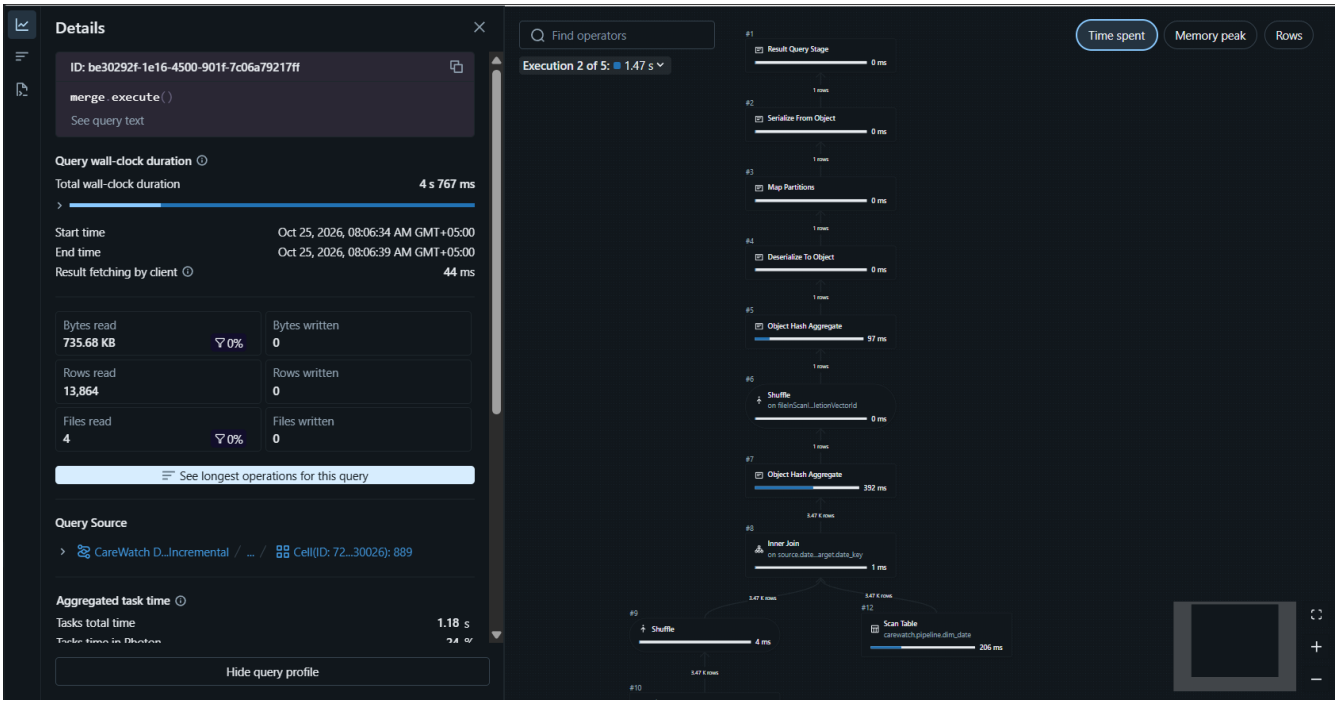


Figure 3. User-supplied query profile for merge.execute(), execution 2 of 5 in Gold task run 450438229000020.

Scope of this screenshot

The visible path reads dim_date, joins on date_key, crosses a Shuffle boundary, and uses Object Hash Aggregate operators. The screenshot is a partial graph of one date-key MERGE. It is not a physical plan for the entire Gold Job or for the largest deficiency fact operation.

The profile does not show enough of the scan predicates or task distribution to prove predicate pushdown or stage skew. Those claims are intentionally excluded.

Statement: be30292f-1e16-4500-901f-7c06a79217ff.

What the observed plan shows

Operator timing and statement metrics give a narrower, defensible reading of Spark execution.

Visible operator	Observed	Interpretation
Scan Table: dim_date	206 ms	The calendar dimension participates in key matching.
Inner Join on date_key	1 ms	A join is visible; its distribution strategy is not established.
Object Hash Aggregate	392 ms; 97 ms	Aggregation occurs in two visible nodes.
Shuffle	4 ms; 0 ms	An exchange boundary is visible in the displayed path.

Statement measurements

4.767 s	13,864	0
wall-clock time	rows read	rows written

The statement read 753,338 bytes from four files. Query history reports 0 spilled bytes and 0 shuffle-read bytes for this statement; these summaries do not erase the visible Shuffle operator.

Scale comparison

A separate fact_deficiency MERGE in the same run processed 418,911 source rows and read 176.1 MB in 7.142 seconds. Its operator screenshot was not supplied, so Figure 3 is not used to describe that larger plan.

Source: Figure 3; system.query.history; Delta history for the scheduled Job.

Observed cost and an hourly scenario

Only the Job charge is tied to this scheduled run. The BI warehouse estimate below is explicitly a planning assumption.

Observed scheduled run	Measured value
Whole Job wall time	489.320 seconds
Serverless Jobs usage	1.359740075 DBU
Effective Jobs list price	USD 0.35 per DBU
Calculated Job list cost	**USD 0.475909**

Hourly enterprise scenario

Repeating the observed Job 8,760 times gives **USD 4,168.96/year** for Jobs compute. An illustrative SQL Import refresh uses 10 active minutes per hour, 0.8 DBU/hour, and USD 0.70/DBU: **USD 817.60/year**. Together these are **USD 4,986.56/year**, before storage, transfer, discounts, taxes, and data growth.

Cost controls

At one daily run, the observed Jobs usage implies USD 173.71/year at list price. Skip unchanged publications, trigger BI only after Gold succeeds, and retain SQL auto-stop. Hourly ingestion may be unnecessary because provider and MDS sources do not necessarily change hourly.

No SQL warehouse usage row was attributable to this Job's 03:00–05:00 UTC window. The USD 0.70 SQL rate is a planning assumption informed by a separate historical hour, not a measured Power BI refresh.

Source: system.billing.usage joined to system.billing.list_prices, 25 October 2026.

Page 1: quality overview

The overview highlights regulatory events and current provider ratings.

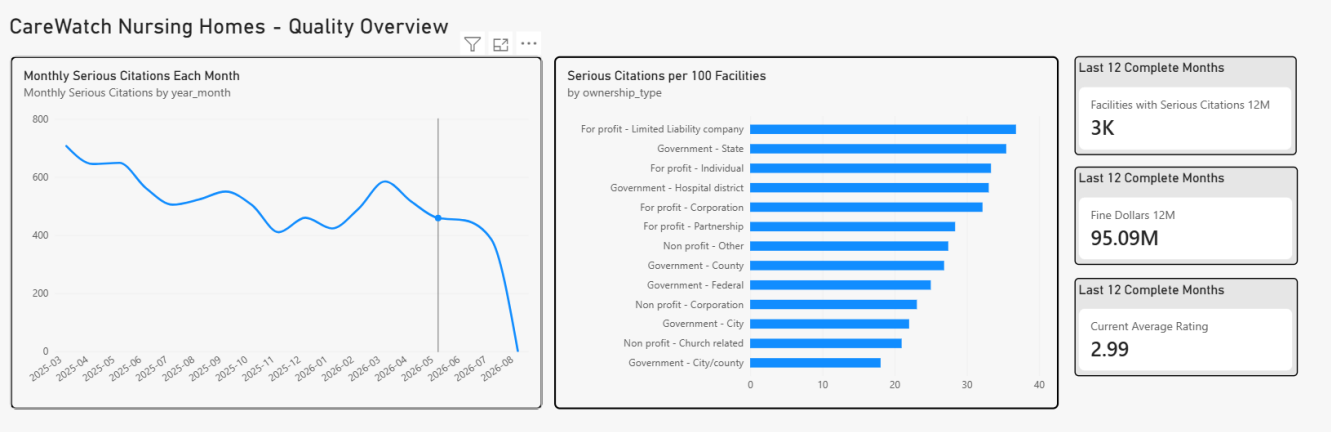


Figure 4. User-supplied Power BI quality overview, 25 October 2026.

4,684
serious citations, 12M

2,933
facilities cited, 12M

$95.09M
fines, 12M

Interpretation

The ownership chart compares serious-citation rates per 100 currently listed facilities. The current average rating is 2.99. These measures help prioritize investigation; they are not a clinical verdict.

Visual checks before submission

The line chart currently spans roughly 18 months and ends with a near-zero August 2026 point. Confirm event completeness and align its filter or title with the 12-month cards. A state map is not visible in this screenshot.

Page 2: facility and staffing

Facility-level penalties and compliance sit beside current staffing and rating.



Figure 5. User-supplied Power BI facility and staffing page.

What this page answers

The ranked bar chart identifies facilities with the largest 12-month fine totals. The matrix allows a reader to inspect citations, fines, and current ratings for a named provider. The staffing visual shows that average reported nurse hours rise across current rating groups.

Interpretation boundary

Current rating incorporates staffing inputs, so this association is not a causal estimate. Use the bar and matrix as a review shortlist, then inspect facility history and source records.

MDS measure context

The selected quality measure currently contains one period-end observation, so the chart is a cross-section.

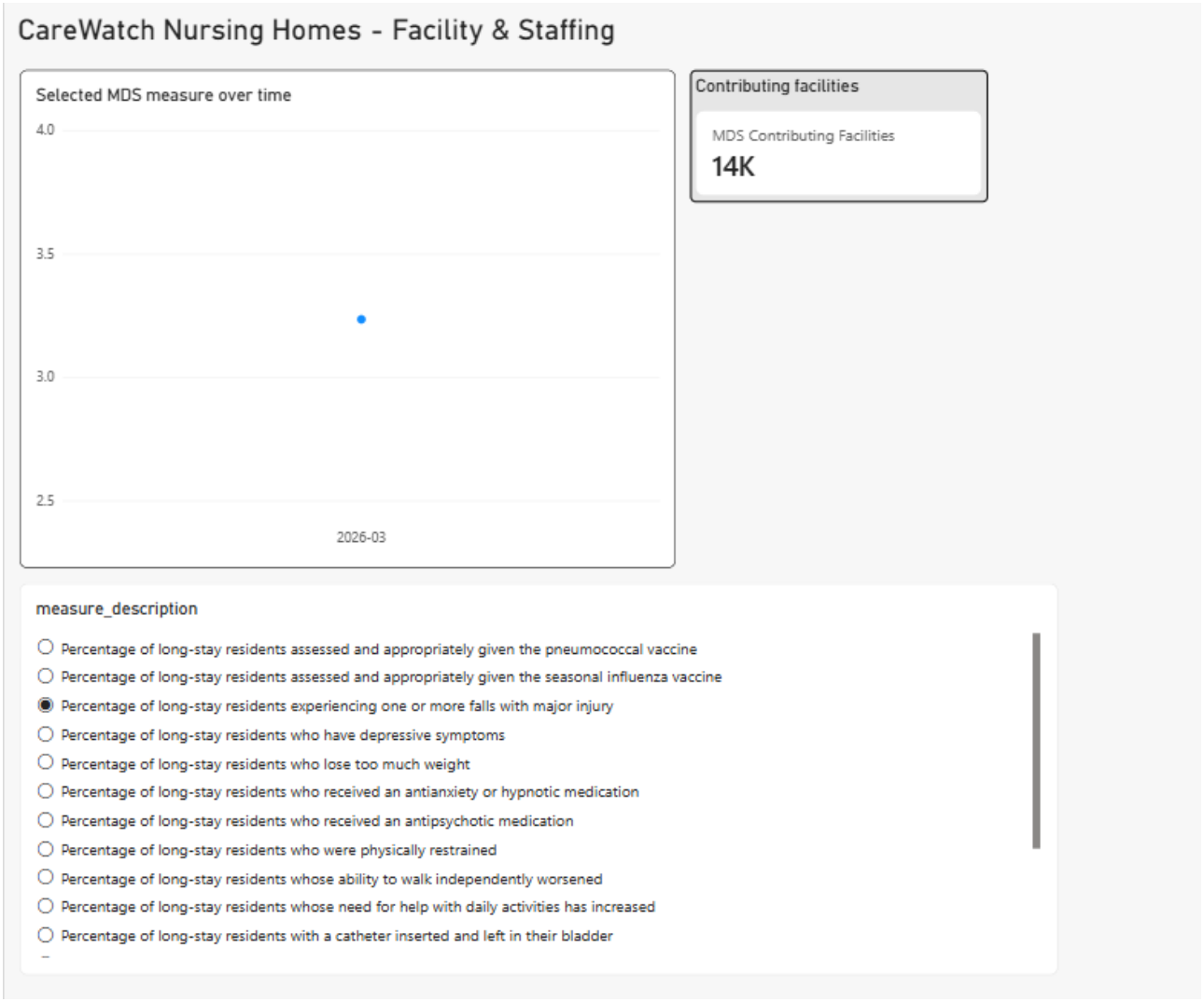


Figure 6. The selected major-injury falls measure has one March 2026 point in the supplied view.

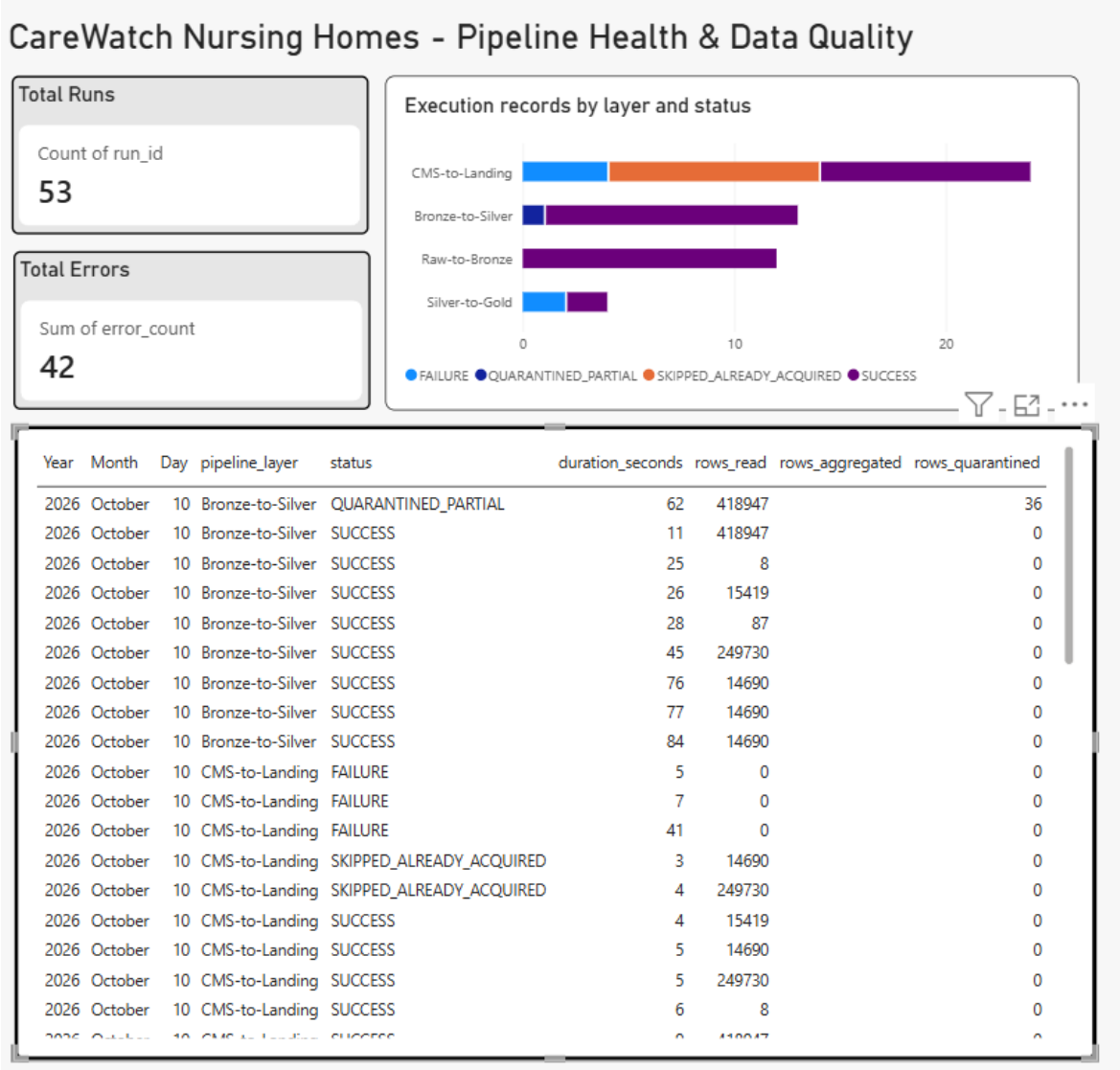
Read the visual precisely

About 14,320 active scored facilities contribute to this selected measure. Its mean is approximately 3.23. With one period, this cannot support a trend claim. The measure selector remains useful for comparing different quality signals when their direction and denominator are explained.

Source: Power BI screenshot and Gold fact_mds_quality / dim_measure model.

Page 3: pipeline health and data quality

The monitoring page reads sanitized Gold audit and error views, separate from the business facts.



Year	Month	Day	pipeline_layer	status	duration_seconds	rows_read	rows_aggregated	rows_quarantined
2026	October	10	Bronze-to-Silver	QUARANTINED_PARTIAL	62	418947		36
2026	October	10	Bronze-to-Silver	SUCCESS	11	418947		0
2026	October	10	Bronze-to-Silver	SUCCESS	25	8		0
2026	October	10	Bronze-to-Silver	SUCCESS	26	15419		0
2026	October	10	Bronze-to-Silver	SUCCESS	28	87		0
2026	October	10	Bronze-to-Silver	SUCCESS	45	249730		0
2026	October	10	Bronze-to-Silver	SUCCESS	76	14690		0
2026	October	10	Bronze-to-Silver	SUCCESS	77	14690		0
2026	October	10	Bronze-to-Silver	SUCCESS	84	14690		0
2026	October	10	CMS-to-Landing	FAILURE	5	0		0
2026	October	10	CMS-to-Landing	FAILURE	7	0		0
2026	October	10	CMS-to-Landing	FAILURE	41	0		0
2026	October	10	CMS-to-Landing	SKIPPED_ALREADY_ACQUIRED	3	14690		0
2026	October	10	CMS-to-Landing	SKIPPED_ALREADY_ACQUIRED	4	249730		0
2026	October	10	CMS-to-Landing	SUCCESS	4	15419		0
2026	October	10	CMS-to-Landing	SUCCESS	5	14690		0
2026	October	10	CMS-to-Landing	SUCCESS	5	249730		0
2026	October	10	CMS-to-Landing	SUCCESS	6	8		0

Figure 7. User-supplied Power BI monitoring page. The capture shows 53 logged execution records and 42 recorded error events.

What the capture shows

A stacked bar compares execution status by pipeline layer. The activity table exposes task duration, rows read, rows aggregated, and rows quarantined. The error card sums aggregated error_count from vw_gold_errors; it does not expose raw error payloads.

Final dashboard check

The independently queried audit view contained 61 records on 25 October, while this Power BI capture displays 53. Reconcile Power BI refresh and filter context before submission. The proposed error-by-rule chart and detail matrix are not visible in this capture.